MATH 241 Spring 2026 Homework #10

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In this problem, we will review the ideas discussed in Section 5.1.

What are some quantities that cannot be modeled by a discrete random variable?
Continuous quantities variables like time, physical measurements, and rates/speeds. True or False? Given a random variable X, it is possible that X does not have a probability mass function. True Define what is meant by a *continuous* random variable. A random variable X is said to be continuous if its Cumulative Distribution Function (CDF), $F(x) = P(X \leq x)$, is continuous everywhere.More practically, X is continuous if there exists a non-negative function $f(x)$ (called the probability density function, or PDF) defined for all real numbers such that for any interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} c(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

What is the value of c? Hint: Apply Humpty-Dumpty.
After using the integration from 0 to 2 , and pulling out the c constant , we solve for the anti derivative and set c * $\frac{8}{3}$=1 and got c =3/8. Suppose that X is a continuous random variable whose probability

density function is given by

$$f(x) = \begin{cases} c(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the cumulative distribution function of X. Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} c(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X]$. Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} c(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X^2]$. Suppose that a random variable X measures the lifetime of a certain type of electronic device in hours. Suppose its probability density function is given by:

$$f(x) = \begin{cases} \frac{10}{x^2}, & \text{if } x > 10 \\ 0, & \text{otherwise} \end{cases}$$

What is the probability that of 6 such types of devices, at least 3 will function for at least 15 hours? (You may assume that the devices operate independently of one another). You may leave your answer in terms of a calculator command.

These exercises will review the king of all random variables – the Normal random variable.

What is the density of the normal random variable? What is the support of the normal random variable? What is the cumulative distribution function of the normal random variable? If $X \sim \mathcal{N}(\mu, \sigma^2)$ then what is $E[X]$? If $X \sim \mathcal{N}(\mu, \sigma^2)$ then what is $Var[X]$? What is the standard normal random variable? What is the density of the standard normal random variable? Use the appropriate letter/notation. What is the cumulative distribution function of the standard normal random variable? Use the appropriate letter/notation. Suppose $Z \sim \mathcal{N}(0, 1)$.

Find $P(Z < 2.41)$. You may leave your answer in terms of an integral, in terms of Φ, in terms of a calculator command, or in terms of an actual number rounded to four decimal places. Suppose $Z \sim \mathcal{N}(0, 1)$.

Find $P(Z > 2.41)$. You may leave your answer in terms of an integral, in terms of Φ, in terms of a calculator command, or in terms of an actual number rounded to four decimal places. Suppose $Z \sim \mathcal{N}(0, 1)$.

Find $P(Z = 2.41)$. Suppose $X \sim \mathcal{N}(241, 17)$. Find $P(X > 231)$. You

may leave your answer in terms of an integral, in terms of Φ, in terms of a calculator command, or in terms of an actual number rounded to four decimal places. Suppose $X \sim \mathcal{N}(241, 17)$. Find $P(X < 231)$. You

may leave your answer in terms of an integral, in terms of Φ, in terms of a calculator command, or in terms of an actual number rounded to four decimal places.

These exercises are designed to review the first half of Section 5.4. Suppose the joint mass function of X and Y is given by:

joint_distribution.png

Define what is meant by a joint mass function (jmf). Find $P(X \leq 2, Y \leq 2)$. Find $f_X(2)$. Find $f_Y(4)$. Find $f_X(x)$.

Find $f_Y(y)$.

What does it mean for two random variables, X and Y, to be independent?

For the joint probability mass function specified above, is X and Y independent? Explain why or why not.

In this problem, we will review the last part of Section 5.4.

Explain what is meant by a convolution of random variables. Suppose

$X_1, X_2 p$. Let $T = X_1 + X_2$. What is the support of T? Find $P(T = 2)$.

Find $P(T = 3)$. Find $P(T = 4)$. Based off the last three computations,

can you guess the probability mass function of T? You do not have to mathematically prove your answer. (If you are curious, the actual proof requires induction and the general convolution formula which you are not required to know).

Here is the last problem for the semester. Congratulations on making it this far! These exercises are designed to review Section 5.5.

Without simply quoting the formal statement of the Central Limit Theorem, can you explain what the Central Limit Theorem states?

Suppose the random variables, $X_1, X_2, \ldots, X_{100}$ are independent and all have the following density:

$$f(x) = \begin{cases} \frac{x^2}{72} & \text{if } 0 < x < 6 \\ 0, & \text{otherwise} \end{cases}$$

If $Y = X_1 + X_2 + \ldots + X_{100}$, then find/approximate $P(439 \leq Y \leq 461)$. You may leave your answer in terms of an integral, in terms of Φ, in terms of a calculator command, or in terms of an actual number rounded to four decimal places.

Suppose the random variables $X_1, X_2, \ldots, X_{36}$ are independent and all have the density specified below.

$$f(x) = \begin{cases} \frac{1}{20}x^3 & \text{if } 1 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

Find/approximate $P(2 \leq \overline{X}_{36} \leq 2.5)$. You may leave your answer in terms of an integral, in terms of Φ, in terms of a calculator command, or in terms of an actual number rounded to four decimal places.